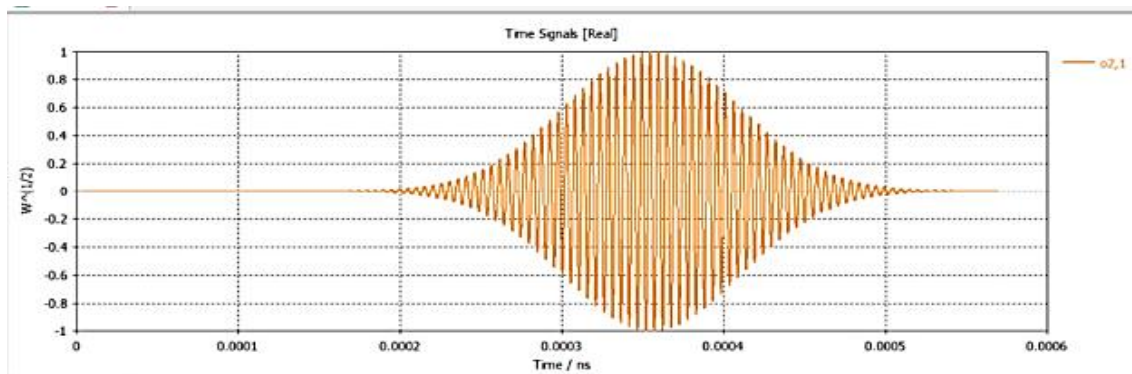
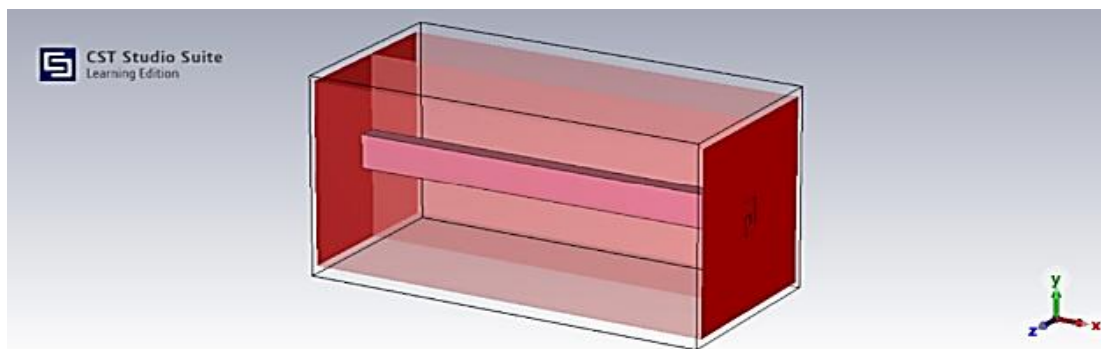


EXNO: 11

Electromagnetic Field and Power Flow Analysis of a Rectangular Waveguide in CST Studio Suite and determination of S parameters.

Test Case 1:

To simulate a Rectangular Waveguide using CST Studio Suite, analyze its electromagnetic behavior, and examine the power distribution among ports over a specified frequency range.



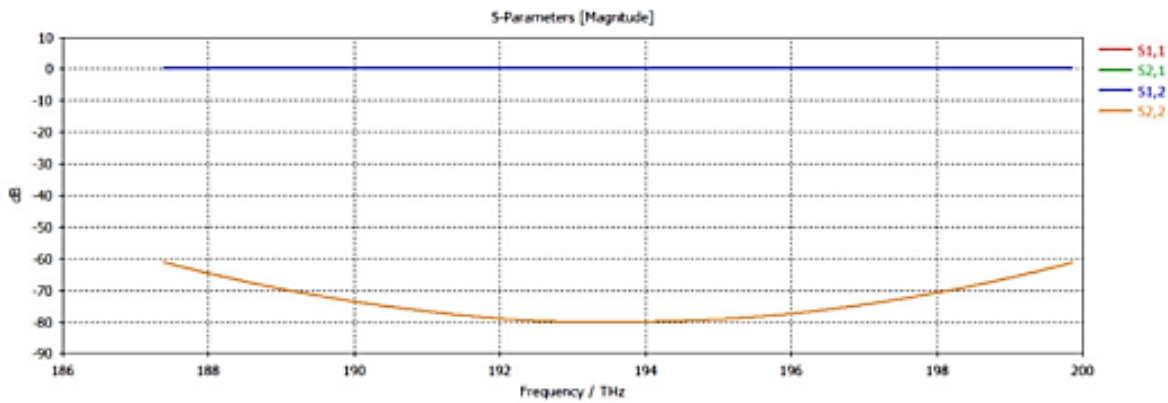
Output:

Port (i-input, o-output)	Time (ns)	Peak Power $W^{1/2}$
O1,1	0.00029	0.016
O1,2	0.00035	1.00
O2,1	0.00035	1.00
O2,2	0.00042	0.016

Test Case 2:

To simulate a Rectangular Waveguide using CST Studio Suite, analyze its electromagnetic behavior, and examine the S parameters.

Simulation Results:



OUTPUT:

Frequency (GHz)	S11 (dB)	S21 (dB)	S12 (dB)	S22 (dB)
187	0	0	0	-62
190	0	0	0	-73
193	0	0	0	-79
196	0	0	0	-76
200	0	0	0	-61